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# Output devices



**Beatriz Sousa Santos, 2026**

Ivan Sutherland's 1965 Vision:

**“The ultimate display”**

**“Don't think of that thing as a screen, think of it as a window, a window through which one looks into a virtual world.**

**The challenge ... is to make that virtual world look real, sound real, move and respond to interaction in real time, and even feel real.”**



# The ultimate display?

"The ultimate display would, of course, be a room within which the computer can control the existence of matter. A chair displayed in such a room would be good enough to sit in. Handcuffs displayed in such a room would be confining, and a bullet displayed in such a room would be fatal." (Ivan Sutherland, 1965)

We are not yet there ... but ...

There are a lot output devices for a lot of different applications  
and evolving ....  
Visual displays:

RATE / MIN 999999  
TOTAL COUNT 999999



**less conventional displays...**





# Graphics/visual Displays

Are computer interfaces that present images to one or several users  
A possible taxonomy:

- Personal displays:
  - monitors
  - HMDs (VR/AR)
  - Monitor-based displays/active glasses
  - Autostereoscopic displays
- Large volume displays:
  - Caves
  - Walls
  - Domes
  - ...

# Personal Displays

The images may be monoscopic or stereoscopic, monocular (for a single eye) or binocular (displayed on both eyes).

- Screens of various sizes
- Wearable Displays
- Hand-held
- Auto-stereoscopic displays  
(desk supported)



# Large-volume displays

- Wall-type displays
- CAVE type displays
- Domes
- ...

[Visual ergonomics: large format screens or multidisplay | GESAB](#)

[Virtual Reality CAVE Solutions | ST Engineering Antycip /](#)





## Main technologies:

- LED displays (several types)
- LCD displays
- Other technologies:
  - autostereoscopic displays: lenticular/barrier
  - electrophoretic...



[E Ink – Wikipedia](#)  
[How Does an E-Ink Screen Work?](#)  
[reMarkable - The future of paper is here | reMarkable](#)



- **Images provided by computer monitors are poor when compared to the real world**
- **It is amazing what we get from such simple devices**
- Monitors have several limitations:
  - Small range of intensities and colors
  - Lack of focusing distance
  - Small field of view
  - ...



# Stereoscopic displays



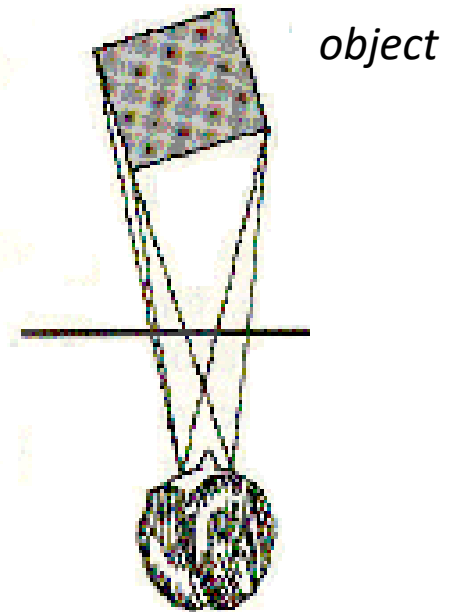
*Two images for the two eyes provided by a HMD  
(Head-Mounted Device)*



*Right eye image    Left eye image*

*Projection plane*

*eyes*



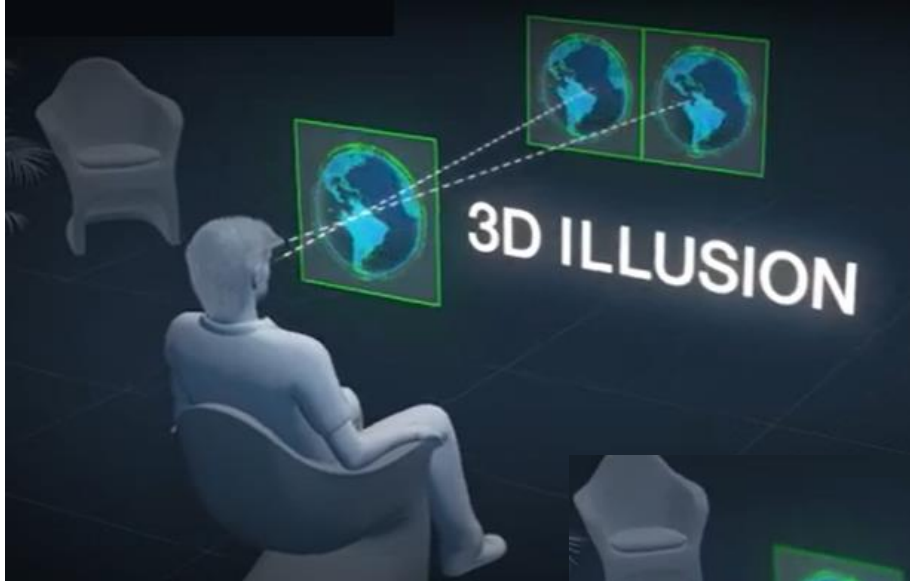
- Need to present **two images** of the same scene (one for the **right eye** and another for the **left eye**)
- The two images can be presented:
  - **at the same time** on two displays (HMD: Head Mounted Display)
  - **time-sequenced** on one display (active glasses)
  - **spatially-sequenced** on one display (auto-stereoscopic displays)



Left eye, right eye images  
(Burdea and Coiffet., 2003)



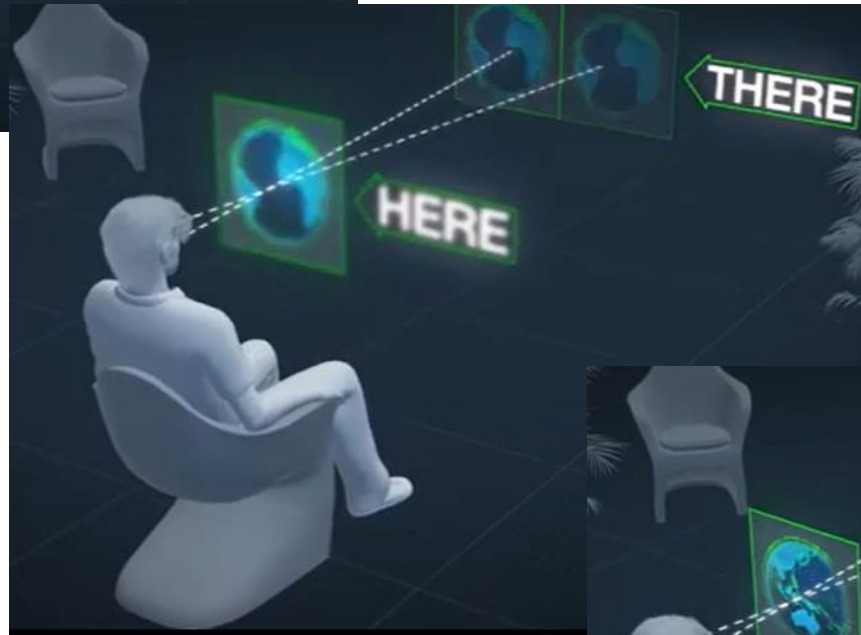




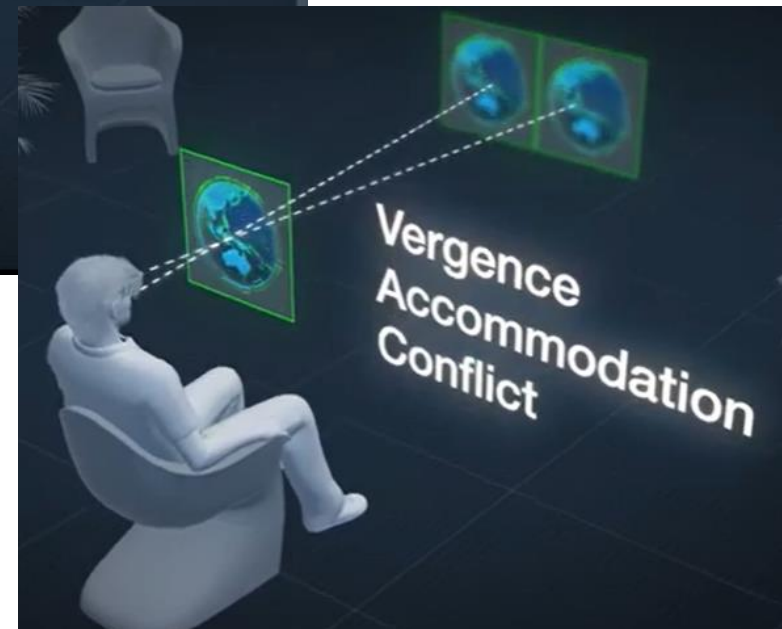
An issue of stereoscopic displays:

Blurred vision, visual discomfort, fatigue, headaches

[Digital Light Shaping™ technology](#)



binocular disparity (vergence) of the 3D rendered content doesn't match the display focal distance (accommodation)



- Other solutions may be used in future  
(3D projections, 3D touch screens, head up displays, ...)



Other solutions for specific cases: HUD – Head-up display already used for some years (first in planes, than in cars)





# Curious about the future of visual displays?



**SolidLight replaces physical things with software-controlled holograms**

[See it Now | Chameleon](#)

[Light Field Lab Demo | ACM SIGGRAPH](#)

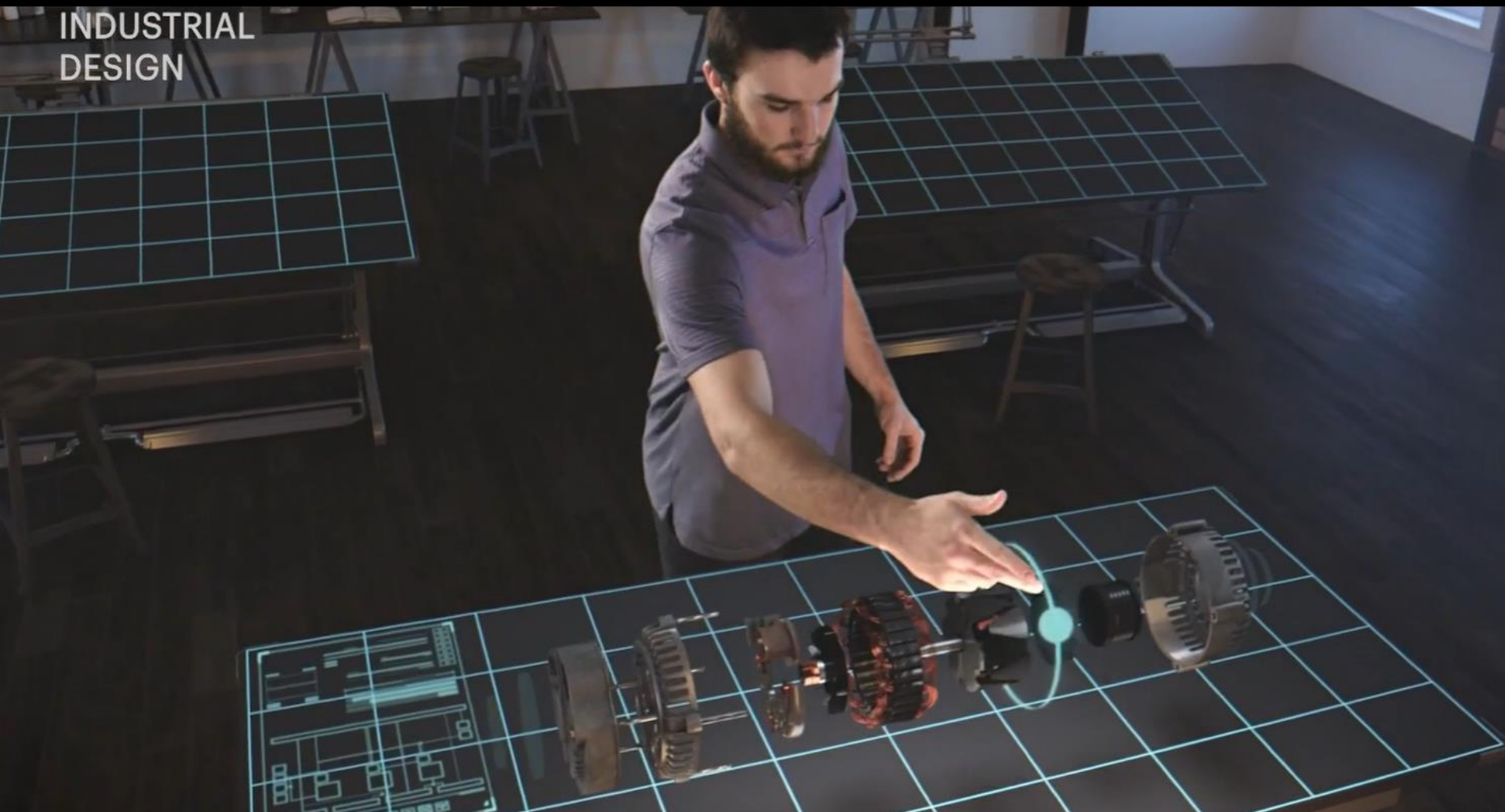
[Augmented reality and virtual reality displays: emerging technologies and future perspectives | Light: Science & Applications](#)

# A glimpse of the future? Interactive live holography

[Light Field Lab](#)

LIGHT FIELD LAB'S VISION FOR THE SOLIDLIGHT™ HOLOGRAPHIC FUTURE

INDUSTRIAL  
DESIGN





**And not only to produce visual displays...**

<https://www.3dsystems.com/haptics-devices/touch>



**sound**



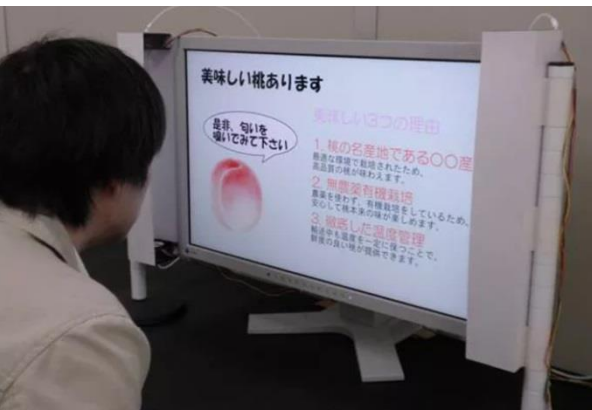
**smell**



**Touch and  
force feedback**

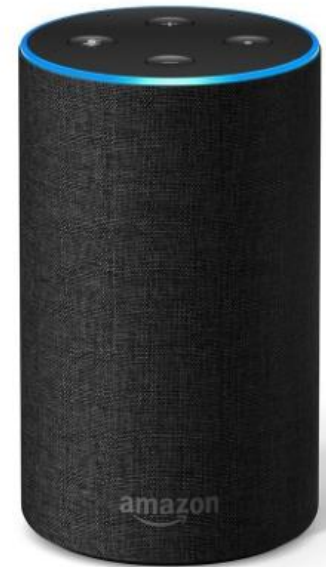


<https://vrscout.com/news/olfactory-engineering-scent-based-vr/>



## Examples of using voice input/output and natural language interaction style:

- Siri
- Alexa
- Google Home
- Google Duplex



[https://en.wikipedia.org/  
wiki/Amazon\\_Alexa](https://en.wikipedia.org/wiki/Amazon_Alexa)

# Voice synthesizers

- There are several types:

Digitized - concatenates recorded basic sounds

Synthesised – concatenates sounds generated with models

- There are several technical challenges due to the nature of human voice:

different pronunciation rules

meaning may be changed by intonation

differences in intonation reflect different moods

- **The quality of a synthesizer implies much more than intelligibility**

## **Advantages of using voice output:**

When the user has:

- physical deficiency
- to move around
- hands and eyes busy
- Adverse conditions: low visibility, low O<sub>2</sub>, high Gs

## **Disadvantages:**

- Is tiresome and uncomfortable for long periods
- Is transient (taxes STM)
- May have privacy issues
- May disturb other people



## Some guidelines to use voice output

- Consider voice output as an alternative when the user must move around, has hands and eyes busy
- Avoid voice output in open environments, when the privacy and security are important issues and frequency of usage is high
- Use approx. 180 words per minute
- When messages are not expected, start with non-critical words that provide context
- Say first the goal and then the solutions
- Allow messages to be repeated

# Every year new devices appear

- Some are really weird ...



## [Haptic System Creates Finger-Touch Sensations Hardware-Free - IEEE Spectrum](#)

- Some never go beyond experimental research
- But once in a while a few become wide-spread

Another example:



**Splashing into a puddle, which triggers a series of short random impulses.**

[Mouth Haptics in VR using a Headset Ultrasound Phased Array | Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems \(acm.org\)](#)

<https://www.youtube.com/watch?v=3q6dZQfV1x8>

# Conclusion

- **Technology shall not be used only because it is new!**
- **Independently from the type or state of the art of the input / output devices it is necessary to understand their usability for different types of users, tasks and context**